Lab Assignemnt-6
Cognitive Computing UCS420
Matplotlib and Seaborn

Q1. Ask the user to enter a value (e.g., M) for a mathematical function.

- Generate x values from -10 to 10 using np.linspace().
- Compute y values for:
  - $y= M \cdot x^2$
  - $y=M \cdot \sin(x)$

- Plot both functions in a single figure:
  - Use different colors and line styles.
  - Add legend, grid, and title.

Q.2 Create a dataset of five subjects and scores.

- Convert it into a Pandas DataFrame.
- Plot the scores using a Seaborn bar plot with:
  - Different colors for each bar.
  - Annotations on top of each bar.
  - Title, axis labels, and grid.

Q3. Write a Python script to select your roll number as seed for NumPy and then generate a dataset of 50 values using np.random.randn().

- Create a 2x2 subplot layout displaying:
  - Line plot showing cumulative sum.
  - Scatter plot with random noise.
- Customize with titles, axis labels, and grids.

Q.4 Download Data-set from the below link
https://github.com/AnjulaMehto/MCA/blob/main/company_sales_data.csv

Apply 'seaborn' library to do the following.

1. Read Total profit of all months and show it using a line plot.
2. Read all product sales data and show it using a multiline plot.
3. Plot bar chart for all the features/attributes.

Lab Assignment 7

UCS420 Cognitive Computing

Q1. You are a data analyst at an e-commerce company. Your task is to analyze sales data, clean and process it, perform statistical analysis, and generate visual insights to help the business make informed decisions.

1. Use your roll number as the random seed to generate unique data for each student.
2. Follow the questions step by step, as each part builds upon the previous one.
3. Submit the assignment as a Jupyter Notebook (.ipynb) with:
    - Well-documented code and clear comments.
    - Markdown explanations for each step.
    - Screenshots of the plots and outputs.

Part I: Randomized Sales Data Generation (NumPy)

1. Initialize Random Seed
2. Generate Sales Data
    - Create a NumPy array of random sales data for 12 months (shape = (12, 4)), where each column represents a product category:
    - Electronics, Clothing, Home & Kitchen, Sports.
    - Sales values should be between 1000 and 5000 units per month.
    - The range should be randomized using the seed.
3. Convert to DataFrame
    - Convert the cleaned NumPy array into a Pandas DataFrame with:
        i. Columns: Electronics, Clothing, Home & Kitchen, Sports
        ii. Index: Months from Jan to Dec

Part II: Data Manipulation and Analysis (Pandas)

1. Display the first 5 rows and summary statistics of the DataFrame.
2. Calculate the total sales per category and total sales per month.
3. Calculate the average sales growth between consecutive months for each category.
4. Add the following columns to the DataFrame:
    - Total Sales: Sum of all categories per month.
    - Growth Rate: Percentage change in Total Sales from the previous month.
5. Apply a randomized discount:
    - If your roll number is even, apply a 10% discount to the Electronics category.

- If odd, apply a 15% discount to the Clothing category.

Part III: Visualizations (Matplotlib and Seaborn)

1. Plot monthly sales trends for each category using line plots.
2. Create the following plots:
    - Box plots to show the sales distribution for each category.

Q.2 For the array: array = np.array([[1, -2, 3],[-4, 5, -6]])

i. Find element-wise absolute value

ii. Find the 25th, 50th, and 75th percentile of flattened array, for each column, for each row.

iii. Mean, Median and Standard Deviation of flattened array, of each column, and each row

Q.3 For the array: a = np.array([-1.8, -1.6, -0.5, 0.5,1.6, 1.8, 3.0]).

i. Find floor, ceiling and truncated value, rounded values

Q.4 WAP to swap two elements in a list in Python Using a Temporary Variable.

Q.5 Write a Python program to swap two elements in a set by converting it to a list first.

Lab Assignemnt-8
Cognitive Computing UCS420
Logistic Regression

1. Implement logistic regression on the Iris dataset.

Lab Assignment 9

UCS420 Cognitive Computing

Assignment Title: NLP using Python

Q1. Write a unique paragraph (5-6 sentences) about your favorite topic (e.g., sports, technology, food, books, etc.).

1. Convert text to lowercase and remove punctuation.
2. Tokenize the text into words and sentences.
3. Remove stopwords (using NLTK's stopwords list).
4. Display word frequency distribution (excluding stopwords).

Q2: Stemming and Lemmatization

1. Take the tokenized words from Question 1 (after stopword removal).
2. Apply stemming using NLTK's PorterStemmer and LancasterStemmer.
3. Apply lemmatization using NLTK's WordNetLemmatizer.
4. Compare and display results of both techniques.

Q3. Regular Expressions and Text Splitting

1. Take their original text from Question 1.
2. Use regular expressions to:
    a. Extract all words with more than 5 letters.
    b. Extract all numbers (if any exist in their text).
    c. Extract all capitalized words.
3. Use text splitting techniques to:
    a. Split the text into words containing only alphabets (removing digits and special characters).
    b. Extract words starting with a vowel.

Q4. Custom Tokenization & Regex-based Text Cleaning

1. Take original text from Question 1.
2. Write a custom tokenization function that:
    a. Removes punctuation and special symbols, but keeps contractions (e.g., "isn't" should not be split into "is" and "n't").
    b. Handles hyphenated words as a single token (e.g., "state-of-the-art" remains a single token).
    c. Tokenizes numbers separately but keeps decimal numbers intact (e.g., "3.14" should remain as is).

3. Use Regex Substitutions (re.sub) to:
    a. Replace email addresses with '<EMAIL>' placeholder.
    b. Replace URLs with '<URL>' placeholder.
    c. Replace phone numbers (formats: 123-456-7890 or +91 9876543210) with '<PHONE>' placeholder.

Lab Assignment 10

UCS420 Cognitive Computing

Assignment Title: NLP using Python-II
(Feature extraction from text, sentiment analysis and text generation)

Q1. Write a unique paragraph (5-6 sentences) about your favorite topic (e.g., sports, technology, food, books, etc.).

1. Convert text to lowercase and remove punctuation using re.
2. Tokenize the text into words and sentences.
3. Split using split() and word_tokenize() and compare how Python split and NLTK's word_tokenize() differ.
4. Remove stopwords (using NLTK's stopwords list).
5. Display word frequency distribution (excluding stopwords).

Q2. Using the same paragraph from Q1:

1. Extract all words with only alphabets using re.findall()
2. Remove stop words using NLTK's stopword list
3. Perform stemming with PorterStemmer
4. Perform lemmatization with WordNetLemmatizer
5. Compare the stemmed and lemmatized outputs and explain when you'd prefer one over the other.

Q3. Choose 3 short texts of your own (e.g., different news headlines, product reviews).

1. Use CountVectorizer to generate the Bag of Words representation.
2. Use TfidfVectorizer to compute TF-IDF scores.
3. Print and interpret the top 3 keywords from each text using TF-IDF.

Q4. Write 2 short texts (4–6 lines each) describing two different technologies (e.g., AI vs Blockchain).

1. Preprocess and tokenize both texts.
2. Calculate:
    a. Jaccard Similarity using sets
    b. Cosine Similarity using TfidfVectorizer + cosine_similarity()

c. Analyze which similarity metric gives better insights in your case.

Q5. Write a short review for a product or service.

1. Use TextBlob or VADER to find polarity & subjectivity for each review.
2. Classify reviews into Positive / Negative / Neutral.
3. Create a word cloud using the wordcloud library for all positive reviews.

Q6. Choose your own paragraph (~100 words) as training data.

1. Tokenize text using Tokenizer() from keras.preprocessing.text
2. Create input sequences and build a simple LSTM or Dense model
3. Train the model and generate 2–3 new lines of text starting from any seed word you provide.